

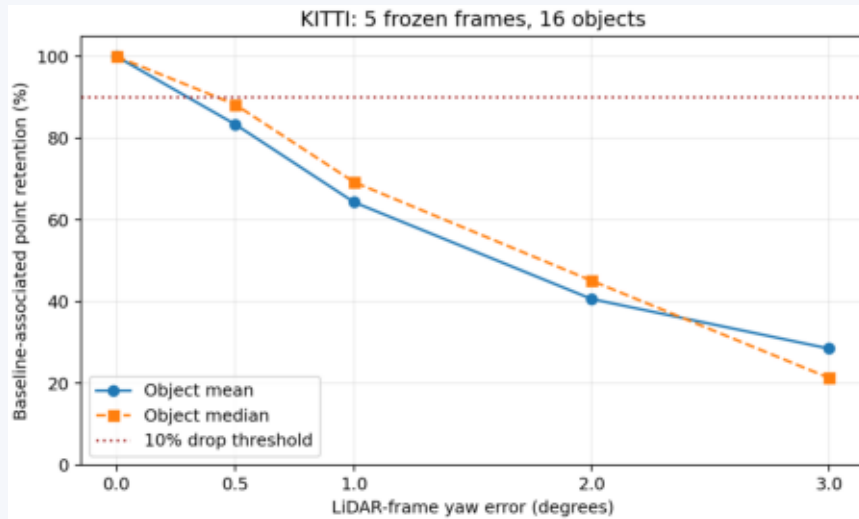
LiDAR-camera calibration drift

Day 6 / Topic A | Dao Gia Bao | 2A202602793

CLAIM SUPPORTED: tested yaw errors $\geq 1^\circ$ cause $\geq 10\%$ relative retention loss.

At 1° : 35.65% loss; mean retention 64.35% (baseline 100%).

01 / Controlled evidence



FROZEN METHOD

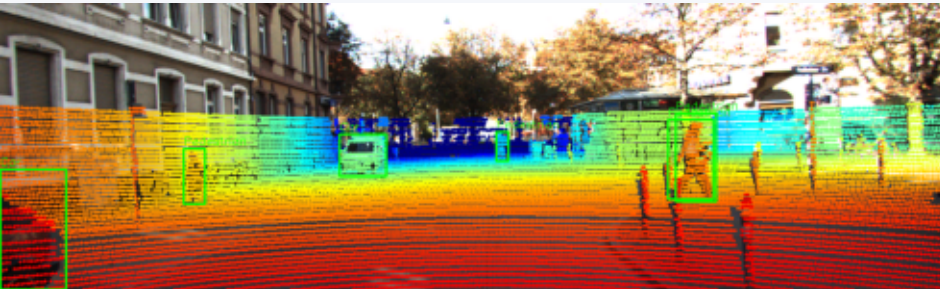
5 KITTI frames / 16 objects / ≥ 5 baseline points
000001, 000004, 000007, 000008, 000009
Car / Pedestrian / Cyclist; seed 42; no sampling
Only yaw changes: 0° , 0.5° , 1° , 2° , 3°
Same point indices; equal weight per object

OTHER MEASUREMENTS

At 3° : mean retention 28.53%
Pixel shift: median 43.58 / p95 60.69 px
FOV stays near 31.4%: weak drift signal
Three CSVs reproduced byte-for-byte in two runs

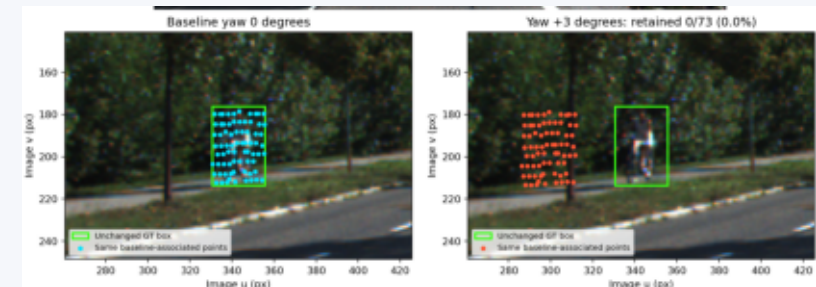
Scope: 2D box association can include background.
Five frames do not establish a universal safety threshold.

02 / Real baseline projection



KITTI 000011: returns overlay people, vehicles and road.
Geometry sanity: (10,0,0) $\rightarrow$ depth 9.727 m, pixel (613.964,175.007).

03 / Geometry failure



Frame 000007, Cyclist #3, yaw + 3° :
0/73 retained; median shift 44.21 px. Same points move outside GT box.

04 / Deployment: ADAS fusion

Monitor persistent alignment residuals and sensor timing; log calibration version, FOV and point counts.
Inspect mounts and recalibrate after drift; reduce fusion confidence until alignment is validated.

Sources: KITTI Vision Benchmark Suite / results/yaw_perturb_sweep.csv. AI: Codex; verified with geometry, repeat runs and figures.